$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#2}_{0^+} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & \mathrm{Y}_1 k^2 + \overset{(2)}{\kappa}_2 & \frac{1}{2}\left(\mathrm{Y}_2 k^2 + 2\,\overset{(2)}{\kappa}_2\right) \\
\mathcal{K}^{\#2}_{0^+}{}^{\dagger} & \frac{1}{2}\left(\mathrm{Y}_2 k^2 + 2\,\overset{(2)}{\kappa}_2\right) & \frac{1}{3}\left(\mathrm{Y}_3 k^2 + 3\,\overset{(2)}{\kappa}_2\right)
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{1}{3}\left(\mathrm{Y}_4 k^2 + \overset{(2)}{\kappa}_2\right) & \frac{1}{6}\left(\sqrt{5}\,\mathrm{Y}_5 k^2 + 2\sqrt{5}\,\overset{(2)}{\kappa}_2\right) \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{1}{6}\left(\sqrt{5}\,\mathrm{Y}_5 k^2 + 2\sqrt{5}\,\overset{(2)}{\kappa}_2\right) & \frac{1}{3}\left(5 k^2\,\overset{(4)}{\kappa}_2 + 5\,\overset{(2)}{\kappa}_2\right)
\end{array}
$$

$$
\begin{array}{c|c}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & \frac{k^2\,\overset{(4)}{\kappa}_3}{3}
\end{array}
\qquad
\begin{array}{c|c}
 & \mathcal{K}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#2}_{0^+} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{\mathrm{Y}_3 k^2 + 3\,\overset{(2)}{\kappa}_2}{3\,\mathrm{Det}(0^+)} & \frac{-\mathrm{Y}_2 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\,\mathrm{Det}(0^+)} \\
\mathcal{J}^{\#2}_{0^+}{}^{\dagger} & \frac{-\mathrm{Y}_2 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\,\mathrm{Det}(0^+)} & \frac{\mathrm{Y}_1 k^2 + \overset{(2)}{\kappa}_2}{\mathrm{Det}(0^+)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{5 k^2\,\overset{(4)}{\kappa}_2 + 5\,\overset{(2)}{\kappa}_2}{3\,\mathrm{Det}(1^-)} & \frac{-\sqrt{5}\,\mathrm{Y}_5 k^2 - 2\sqrt{5}\,\overset{(2)}{\kappa}_2}{6\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{-\sqrt{5}\,\mathrm{Y}_5 k^2 - 2\sqrt{5}\,\overset{(2)}{\kappa}_2}{6\,\mathrm{Det}(1^-)} & \frac{\mathrm{Y}_4 k^2 + \overset{(2)}{\kappa}_2}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|c}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & \frac{3}{k^2\,\overset{(4)}{\kappa}_3}
\end{array}
\qquad
\begin{array}{c|c}
 & \mathcal{J}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & 0
\end{array}
$$

Abbreviations used in matrices

$$\mathrm{Y}_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&\; \mathrm{Y}_2 == 2\left(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2\right) + \overset{(4)}{\kappa}_4 \;\&\&$$

$$\mathrm{Y}_3 == 3\left(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2\right) + \overset{(4)}{\kappa}_3 \;\&\&\; \mathrm{Y}_4 == \overset{(4)}{\kappa}_2 + 2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&\; \mathrm{Y}_5 == 2\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_4 \;\&\&$$

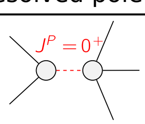
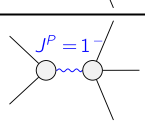
$$\mathrm{Det}(0^+) == \frac{1}{12}\left(k^4\left(4\,\overset{(4)}{\kappa}_3\left(4\left(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2\right) + \overset{(4)}{\kappa}_3\right) + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 - 3\,\overset{(4)}{\kappa}_4{}^2\right) + 16\,k^2\,\overset{(4)}{\kappa}_3\,\overset{(2)}{\kappa}_2\right) \;\&\&$$

$$\mathrm{Det}(1^-) == \frac{5}{36}\,k^2\left(k^2\left(8\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4{}^2\right) + 8\,\overset{(4)}{\kappa}_3\,\overset{(2)}{\kappa}_2\right)$$

Lagrangian

$$\overset{(2)}{\kappa}_2\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}_{\beta}{}^{\chi}{}_{\chi} + \overset{(4)}{\kappa}_1\,\partial_{\beta}\mathcal{K}_{\chi}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_{\chi}\mathcal{K}_{\beta}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta} + \overset{(4)}{\kappa}_4\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{3^-}{}^{\alpha\beta\chi} == 0$	7	$k^{\delta}\,k^{\epsilon}\,(3\,\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha\phi}{}_{\phi} + 2\,\partial_{\phi}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta\epsilon}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha}{}_{\delta}{}^{\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\phi} - 5\,\partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} + \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\mathcal{J}_{\delta}{}^{\phi\gamma} - \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi\gamma} - \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi\gamma} - \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi\gamma} + \eta^{\beta\chi}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi}{}_{\phi} + \eta^{\alpha\chi}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi}{}_{\phi} + \eta^{\alpha\beta}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi}{}_{\phi}) == 0$
Total # constraint(s):	7	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{16\,\overset{(4)}{\kappa}_3\,\overset{(2)}{\kappa}_2}{4\,\overset{(4)}{\kappa}_3\left(4\left(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2\right)+\overset{(4)}{\kappa}_3\right)+4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4-3\,\overset{(4)}{\kappa}_4{}^2}$	$\frac{20\,\overset{(4)}{\kappa}_3{}^2+12\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4+9\,\overset{(4)}{\kappa}_4{}^2}{32\,\overset{(4)}{\kappa}_1\,\overset{(4)}{\kappa}_3{}^2+32\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3{}^2+8\,\overset{(4)}{\kappa}_3{}^3+8\,\overset{(4)}{\kappa}_3{}^2\,\overset{(4)}{\kappa}_4-6\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4{}^2}$
	3	$\frac{8\,\overset{(4)}{\kappa}_3\,\overset{(2)}{\kappa}_2}{-8\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\overset{(4)}{\kappa}_4{}^2}$	$-\frac{24\,\overset{(4)}{\kappa}_3{}^2+12\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4+9\,\overset{(4)}{\kappa}_4{}^2}{40\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3{}^2-5\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4{}^2}$
Resolved unitarity condition(s):	(Demonstrably impossible)		